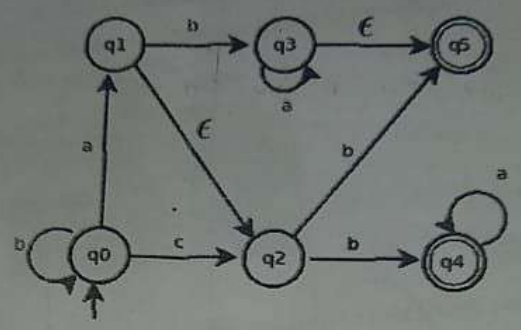
Universidade Estadual de Mato Grosso do Sul - UEMS
2º ano de Ciência da Computação
2ª Avaliação de Linguagens Formais e Autômatos - 26/08/2024
Professor Osvaldo Vargas Jaques

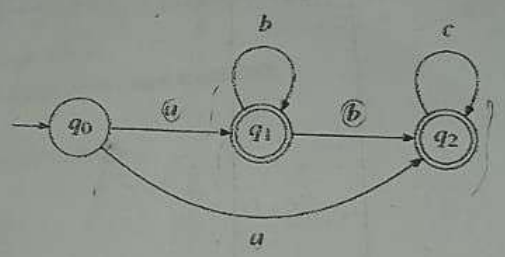
ALUNO: Felipe Eduardo Vilhalva RGM: 45611

1. (2,0 pontos) Minimize o seguinte autômato finito representado abaixo:



2. (2,0 pontos) Construa um automato finito que aceite a linguagem gerada pela expressão regular $ab^*abb^* \cup (a^* \cup b^*a)bbb^*$, definida sobre o alfabeto a, b .

3. (2,0 pontos) Encontre a expressão regular correspondente ao AF abaixo:



4. (2,0 pontos) Sobre o Teorema do Bombeamento para as linguagens regulares:
   a) O que é?
   b) Qual o seu enunciado?
   c) Quais são suas hipóteses?
   d) No que se baseia sua demonstração?
   e) Qual sua principal aplicação?
   f) Em quais passos pode ser decomposta essa sua principal aplicação?

5. (2,0 pontos) Prove que a seguinte linguagem $\{a^i b^i | i \geq 1\}$ não é regular.

"Ninguém, depois de acender uma candeia, a põe em lugar escondido, nem debaixo do alquire, mas no velador, a fim de que os que entram vejam a luz."Lucas 11:33
De nada vale o maior conhecimento que uma pessoa possa ter se ela não passá-lo adiante."

Aluno: Henrique G. Oliveira ; RGM: 45767

① → 1º converter o AFD

$E(0) = \{q_0\}$ ⊕

$\{q_0\}$:
- $E(\delta(\{q_0\},a)) = E(\{q_1\}) = \{q_1,q_2\}$ ⊕
- $E(\delta(\{q_0\},b)) = E(\{q_0\}) = \{q_0\}$ -
- $E(\delta(\{q_0\},c)) = E(\{q_2\}) = \{q_2\}$ ⊕

$\{q_1,q_2\}$:
- $E(\delta(\{q_1,q_2\},a)) = E(\{\}) = \{\}$ ⊕
- $E(\delta(\{q_1,q_2\},b)) = E(\{q_3,q_4,q_5\}) = \{q_3,q_4,q_5\}$ ⊕
- $E(\delta(\{q_1,q_2\},c)) = E(\{\}) = \{\}$ -

$\{q_2\}$:
- $E(\delta(\{q_2\},a)) = E(\{\}) = \{\}$ -
- $E(\delta(\{q_2\},b)) = E(\{q_4,q_5\}) = \{q_4,q_5\}$ ⊕
- $E(\delta(\{q_2\},c)) = E(\{\}) = \{\}$ -

$\{\}$:
- $E(\delta(\{\},a)) = E(\{\}) = \{\}$ -
- $E(\delta(\{\},b)) = E(\{\}) = \{\}$ -
- $E(\delta(\{\},c)) = E(\{\}) = \{\}$ -

$\{q_3,q_4,q_5\}$:
- $E(\delta(\{q_3,q_4,q_5\},a)) = E(\{q_3,q_4\}) = \{q_3,q_4,q_5\}$ ✓
- $E(\delta(\{q_3,q_4,q_5\},b)) = E(\{\}) = \{\}$ -
- $E(\delta(\{q_3,q_4,q_5\},c)) = E(\{\}) = \{\}$ -

$\{q_4,q_5\}$:
- $E(\delta(\{q_4,q_5\},a)) = E(\{q_4\}) = \{q_4\}$ ⊕
- $E(\delta(\{q_4,q_5\},b)) = E(\{\}) = \{\}$ -
- $E(\delta(\{q_4,q_5\},c)) = E(\{\}) = \{\}$ -

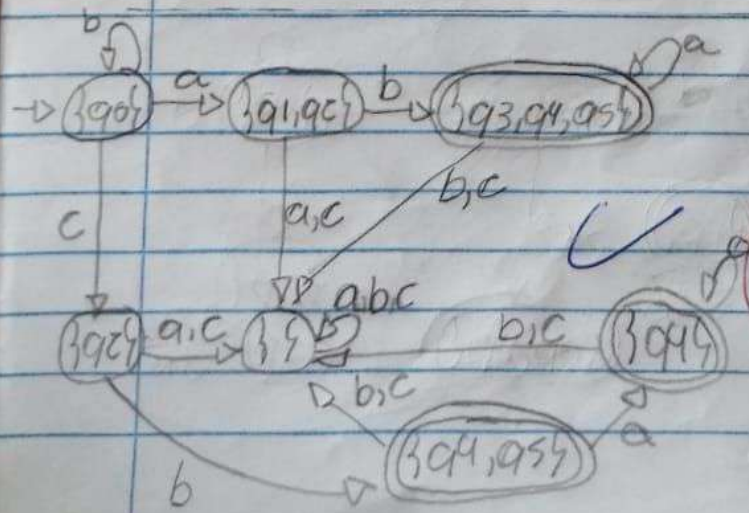
$\{q_4\}$:
- $E(\delta(\{q_4\},a)) = E(\{q_4\}) = \{q_4\}$ ✓
- $E(\delta(\{q_4\},b)) = E(\{\}) = \{\}$ -
- $E(\delta(\{q_4\},c)) = E(\{\}) = \{\}$ -

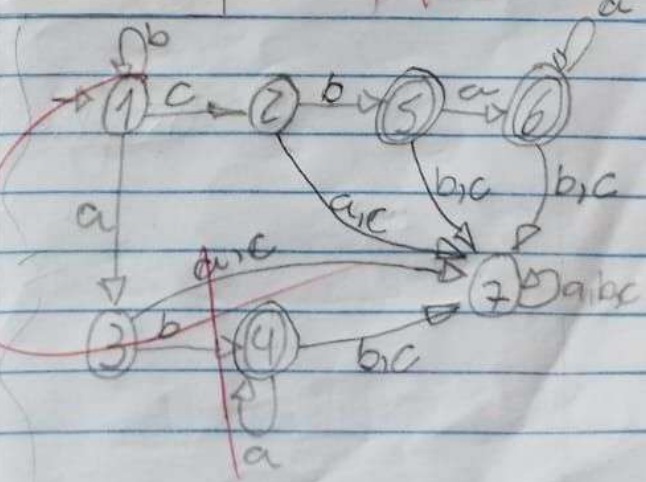
estados D	Marca
$\{q_0\}$	✓
$\{q_1,q_2\}$	✓
$\{q_2\}$	✓
$\{\}$	✓
$\{q_3,q_4,q_5\}$	✓
$\{q_4,q_5\}$	✓
$\{q_4\}$	✓

6

$\{q_0\} \Rightarrow 1$
$\{q_2\} \Rightarrow 2$
$\{q_1,q_2\} \Rightarrow 3$
$\{q_3,q_4,q_5\} \Rightarrow 4$
$\{q_4,q_5\} \Rightarrow 5$
$\{q_4\} \Rightarrow 6$
$\{\} \Rightarrow 7$

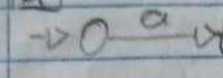
1,0 5





(c) $ab^*abb^* \cup (a^* \cup b^*a)bbb^*$

$\boxed{a}$

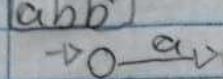


;

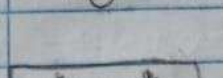
$\boxed{b}$

;

$\boxed{ab^*}$

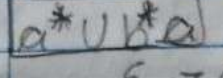


$\boxed{abb^*}$

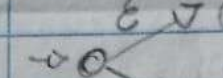


;

$\boxed{ab^*abb^*}$

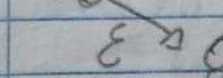


$\boxed{a^* \cup b^*a}$

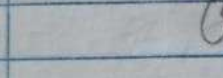


;

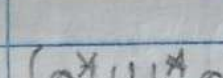
$\boxed{bbb^*}$



$(a^* \cup b^*a)bbb^*$



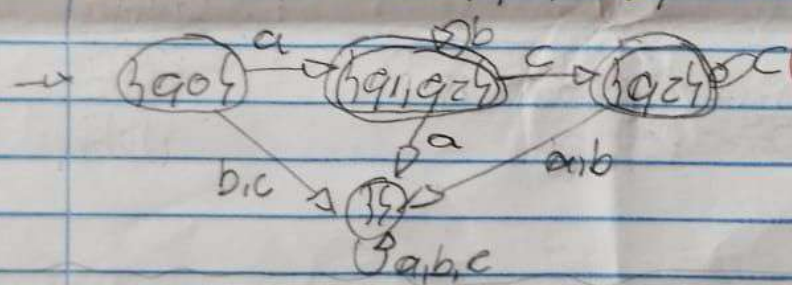
$ab^*abb^* \cup (a^* \cup b^*a)bbb^*$

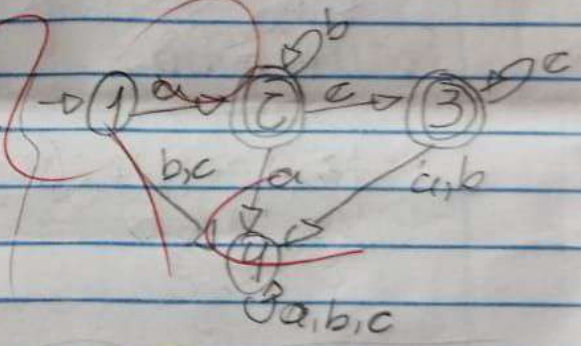


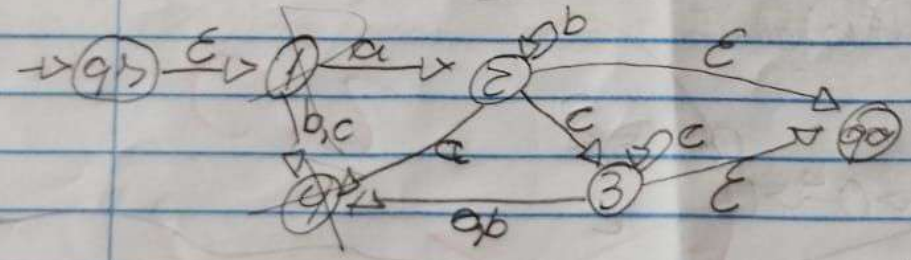
③ → transformar p/ AFD

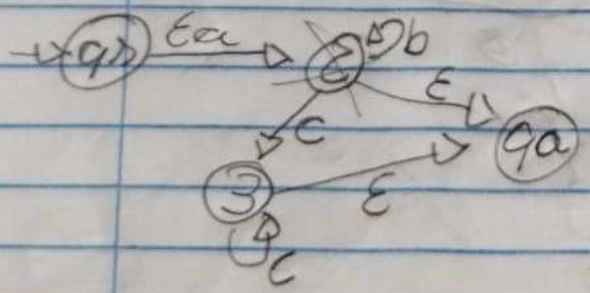
	estados D	Marca
$E(0) = \{q_0\}$ ⊗	$\{q_0\}$	✓
$E(\delta(\{q_0\}, a)) = E(\{q_1, q_2\}) = \{q_1, q_2\}$ ⊗	$\{q_1, q_2\}$	✓
$E(\delta(\{q_0\}, b)) = E(\{\}) = \{\}$ ⊗	$\{\}$	✓
$E(\delta(\{q_0\}, c)) = E(\{\}) = \{\}$	$\{q_2\}$	✓

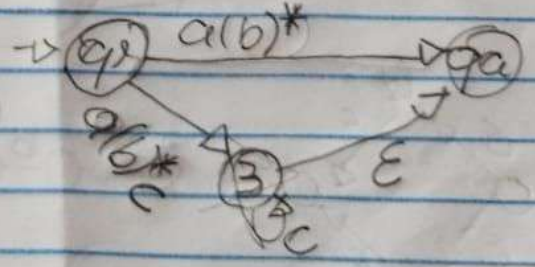
$E(\delta(\{\}, a)) = E(\{\}) = \{\}$
$E(\delta(\{\}, b)) = E(\{\}) = \{\}$
$E(\delta(\{\}, c)) = E(\{\}) = \{\}$
$E(\delta(\{q_1, q_2\}, a)) = E(\{\}) = \{\}$
$E(\delta(\{q_1, q_2\}, b)) = E(\{q_1, q_2\}) = \{q_1, q_2\}$
$E(\delta(\{q_1, q_2\}, c)) = E(\{q_2\}) = \{q_2\}$
$E(\delta(\{q_2\}, a)) = E(\{\}) = \{\}$
$E(\delta(\{q_2\}, b)) = E(\{\}) = \{\}$
$E(\delta(\{q_2\}, c)) = E(\{q_2\}) = \{q_2\}$











→ q1 $(a(b)^*) \cup (a(b)^*c(c)^*)$ → qa

⇒ $(a(b)^*) \cup (a(b)^*c(c)^*)$